

Tarea 2: Instalación de Docker

1. Entramos al repositorio de ubuntu

```
ubuntu@ip-172-31-25-3: ~
#
alumnado@leppsur:~$ ls
'Desarrollo Web en Entorno Cliente'  'Despliegue Web'  Escritorio  Música  PYTHON  Videos
'Desarrollo Web Entorno'             Documentos        Imágenes   Plantillas  Repositorio1
Descargas                             eclipse-workspace 'Interfaces web'  Público    snap
alumnado@leppsur:~$ cd Despliegue Web/
alumnado@leppsur:~/Despliegue Web$ chmod 400 "daweb2025.pem"
alumnado@leppsur:~/Despliegue Web$ ssh -l "daweb2025.pem" ubuntu@ec2-3-85-134-37.compute-1.amazonaws.com
The authenticity of host 'ec2-3-85-134-37.compute-1.amazonaws.com (3.85.134.37)' can't be established.
ED25519 key fingerprint is SHA256:9ddvKBGh20ndZVAKytfnS5tTV50p1EZxSRWshT/qE80.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-85-134-37.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 26 11:08:08 UTC 2025

System load:  0.01               Temperature:  -273.1 C
Usage of /:   25.6% of 6.71GB    Processes:    115
Memory usage: 23%              Users logged in: 0
Swap usage:   0%               IPv4 address for ens5: 172.31.25.3

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

2. Dentro del repositorio de ubuntu, preparamos la instalación de docker, lo cual, contará con varios pasos que mostraré a continuación.

```
ubuntu@ip-172-31-25-3: ~
#
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-25-3:~$ sudo apt update
Hit1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1171 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [381 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1443 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [282 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.3 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1485 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [299 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [377 kB]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.0 kB]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1957 kB]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [441 kB]
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [1212 B]
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [544 B]
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.0 kB]
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [5500 B]
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:29 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [488 B]
Get:30 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [39.9 kB]
Get:31 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9152 B]
Get:32 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7072 B]
Get:33 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [272 B]
Get:34 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.2 kB]
Get:35 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [17.4 kB]
Get:36 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [19.2 kB]
Get:37 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1304 B]
```

```
ubuntu@ip-172-31-25-3: ~
#
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
38 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-25-3:~$ sudo apt install apt-transport-https ca-certificates curl software-properties-common
Reading package lists... Done
Building dependency tree... Done
```

```
ubuntu@ip-172-31-25-3:~$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo apt-key add -
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (see apt-key(8)).
OK
```

```
ubuntu@ip-172-31-25-3:~$ sudo add-apt-repository "deb [arch=amd64] https://download.docker.com/linux/ubuntu focal stable"
Repository: 'deb [arch=amd64] https://download.docker.com/linux/ubuntu focal stable'
Description:
Archive for codename: focal components: stable
More info: https://download.docker.com/linux/ubuntu
Adding repository.
Press [ENTER] to continue or Ctrl-c to cancel.
Adding deb entry to /etc/apt/sources.list.d/archive_uri-https_download_docker_com_linux_ubuntu-noble.list
Adding disabled deb-src entry to /etc/apt/sources.list.d/archive_uri-https_download_docker_com_linux_ubuntu-noble.list
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 https://download.docker.com/linux/ubuntu focal InRelease [57.7 kB]
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 https://download.docker.com/linux/ubuntu focal/stable amd64 Packages [59.9 kB]
Fetched 118 kB in 1s (185 kB/s)
Reading package lists... Done
W: https://download.docker.com/linux/ubuntu/dists/focal/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
```

```
ubuntu@ip-172-31-25-3:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
38 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

3. Instalamos docker

```
ubuntu@ip-172-31-25-3: ~
ubuntu@ip-172-31-25-3: ~ 122x40
ubuntu@ip-172-31-25-3:~$ sudo apt install docker-ce
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  containerd.io docker-buildx-plugin docker-ce-cli docker-ce-rootless-extras docker-compose-plugin libltdl7 libslirp0
  pigz slirp4netns
Suggested packages:
  cgroupfs-mount | cgroup-lite
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin libltdl7
  libslirp0 pigz slirp4netns
0 upgraded, 10 newly installed, 0 to remove and 38 not upgraded.
Need to get 120 MB of archives.
After this operation, 440 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libltdl7 amd64 2.4.7-7build1 [40.3 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libslirp0 amd64 4.7.0-1ubuntu3 [63.8 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 slirp4netns amd64 1.2.1-1build2 [34.9 kB]
Get:5 https://download.docker.com/linux/ubuntu focal/stable amd64 containerd.io amd64 1.7.27-1 [30.5 MB]
Get:6 https://download.docker.com/linux/ubuntu focal/stable amd64 docker-buildx-plugin amd64 0.23.0-1-ubuntu.20.04-focal [
34.6 MB]
```

4. Comprobamos que funciona

```
ubuntu@ip-172-31-25-3:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-09-26 11:18:00 UTC; 1min 23s ago
 TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 3650 (dockerd)
      Tasks: 9
  Memory: 84.3M (peak: 84.6M)
     CPU: 504ms
    CGroup: /system.slice/docker.service
            └─3650 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Sep 26 11:17:58 ip-172-31-25-3 dockerd[3650]: time="2025-09-26T11:17:58.252578523Z" level=info msg="detected 127.0.0.53 n
Sep 26 11:17:59 ip-172-31-25-3 dockerd[3650]: time="2025-09-26T11:17:59.112036297Z" level=info msg="Creating a containerd
Sep 26 11:17:59 ip-172-31-25-3 dockerd[3650]: time="2025-09-26T11:17:59.231088235Z" level=info msg="Loading containers: s
Sep 26 11:18:00 ip-172-31-25-3 dockerd[3650]: time="2025-09-26T11:18:00.008771290Z" level=info msg="Loading containers: d
Sep 26 11:18:00 ip-172-31-25-3 dockerd[3650]: time="2025-09-26T11:18:00.117987666Z" level=info msg="Docker daemon" commit
Sep 26 11:18:00 ip-172-31-25-3 dockerd[3650]: time="2025-09-26T11:18:00.118121294Z" level=info msg="Initializing buildkit"
Sep 26 11:18:00 ip-172-31-25-3 dockerd[3650]: time="2025-09-26T11:18:00.209632984Z" level=info msg="Completed buildkit ini
Sep 26 11:18:00 ip-172-31-25-3 dockerd[3650]: time="2025-09-26T11:18:00.221694673Z" level=info msg="Daemon has completed
Sep 26 11:18:00 ip-172-31-25-3 dockerd[3650]: time="2025-09-26T11:18:00.221933216Z" level=info msg="API listen on /run/doc
Sep 26 11:18:00 ip-172-31-25-3 systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)
```

5. Por defecto, el comando docker solo puede ser ejecutado por el usuario root o un usuario del grupo docker nos metemos en el grupo del Docker para poder manipularlo.

```
ubuntu@ip-172-31-25-3:~$ sudo usermod -aG docker ${USER}
ubuntu@ip-172-31-25-3:~$
```

6. Para hacer efectivo el cambio hay que reiniciar el docker

```
ubuntu@ip-172-31-25-3: ~
ubuntu@ip-172-31-25-3: ~ 122x40

ubuntu@ip-172-31-25-3:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
ubuntu@ip-172-31-25-3:~$ id -nG
ubuntu adm cdrom sudo dip lxd docker
ubuntu@ip-172-31-25-3:~$ service docker status
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-09-26 11:30:14 UTC; 2min 34s ago
 TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 730 (dockerd)
      Tasks: 9
  Memory: 93.6M (peak: 94.0M)
     CPU: 584ms
    CGroup: /system.slice/docker.service
            └─730 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Sep 26 11:30:13 ip-172-31-25-3 dockerd[730]: time="2025-09-26T11:30:13.166204143Z" level=info msg="[graphdriver] using pr
Sep 26 11:30:13 ip-172-31-25-3 dockerd[730]: time="2025-09-26T11:30:13.170303807Z" level=info msg="Loading containers: st
Sep 26 11:30:13 ip-172-31-25-3 dockerd[730]: time="2025-09-26T11:30:13.841612100Z" level=warning msg="Error (Unable to co
Sep 26 11:30:13 ip-172-31-25-3 dockerd[730]: time="2025-09-26T11:30:13.963906714Z" level=info msg="Loading containers: do
Sep 26 11:30:14 ip-172-31-25-3 dockerd[730]: time="2025-09-26T11:30:14.046130268Z" level=info msg="Docker daemon" commit
Sep 26 11:30:14 ip-172-31-25-3 dockerd[730]: time="2025-09-26T11:30:14.047010221Z" level=info msg="Initializing buildkit"
Sep 26 11:30:14 ip-172-31-25-3 dockerd[730]: time="2025-09-26T11:30:14.090830842Z" level=info msg="Completed buildkit ini
Sep 26 11:30:14 ip-172-31-25-3 dockerd[730]: time="2025-09-26T11:30:14.124318201Z" level=info msg="Daemon has completed i
Sep 26 11:30:14 ip-172-31-25-3 systemd[1]: Started docker.service - Docker Application Container Engine.
Sep 26 11:30:14 ip-172-31-25-3 dockerd[730]: time="2025-09-26T11:30:14.124637710Z" level=info msg="API listen on /run/doc

ubuntu@ip-172-31-25-3:~$
```

7. Ahora pasamos a la instalación de Apache, para eso actualizamos los paquetes

```
ubuntu@ip-172-31-25-3:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 https://download.docker.com/linux/ubuntu focal InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
38 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: https://download.docker.com/linux/ubuntu/dists/focal/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
ubuntu@ip-172-31-25-3:~$
```

8. Instalamos Apache

```
ubuntu@ip-172-31-25-3:~$ sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0
  ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 38 not upgraded.
Need to get 2086 kB of archives.
After this operation, 8090 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2
kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116 B]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.8 [1331 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.8 [163 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.8 [97.7
kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.8 [90.2 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
```

9. Dentro del docker volvemos a instalar apache

```
ubuntu@lp-172-31-25-3:~$ docker exec -it prueba2 /bin/bash
root@14a14a437a1:/# apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [5609 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3371 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1274 kB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [71.0 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [266 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [164 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages [1792 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [17.5 MB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1577 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3690 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [69.1 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [5804 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [35.2 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [83.2 kB]
Fetched 41.9 MB in 3s (13.6 MB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@14a14a437a1:/#
```

```
root@14a14a437a1:/# apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils bzip2 ca-certificates file libapr1 libaprutil1 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libbrotli1 libcurl4 libexpat1 libgdbm-compat4 libgdbm6 libicu70 libjansson4 libldap-2.5-0
  libldap-common liblua5.3-0 libmagic-mgc libmagic1 libnghttp2-14 libperl5.34 libpsl5 librtmp1 libssl2-2
  libsasl2-modules libsasl2-modules-db libsasl2-modules-gssapi-mit libsasl2-modules-gssapi-heimdal libsasl2-modules-ldap libsasl2-modules-otp
  libsasl2-modules-sql perl perl-modules-5.34 publicsuffix ssl-cert xz-utils
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser ufw bzip2-doc gdbm-l10n
  libsasl2-modules-gssapi-mit | libsasl2-modules-gssapi-heimdal libsasl2-modules-ldap libsasl2-modules-otp
  libsasl2-modules-sql perl-doc libterm-readline-gnu-perl | libterm-readline-perl-perl make libtap-harness-archive-perl
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils bzip2 ca-certificates file libapr1 libaprutil1 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libbrotli1 libcurl4 libexpat1 libgdbm-compat4 libgdbm6 libicu70 libjansson4 libldap-2.5-0
  libldap-common liblua5.3-0 libmagic-mgc libmagic1 libnghttp2-14 libperl5.34 libpsl5 librtmp1 libssl2-2
  libsasl2-modules libsasl2-modules-db libsasl2-modules-gssapi-mit libsasl2-modules-gssapi-heimdal libsasl2-modules-ldap
  libsasl2-modules-otp libsasl2-modules-sql perl perl-modules-5.34 publicsuffix ssl-cert xz-utils
0 upgraded, 43 newly installed, 0 to remove and 3 not upgraded.
Need to get 25.6 MB of archives.
After this operation, 111 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 perl-modules-5.34 all 5.34.0-3ubuntu1.5 [2977 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 libgdbm6 amd64 1.23-1 [33.9 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 libgdbm-compat4 amd64 1.23-1 [6606 B]
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libperl5.34 amd64 5.34.0-3ubuntu1.5 [4797 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 perl amd64 5.34.0-3ubuntu1.5 [232 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [108 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libexpat1 amd64 2.4.7-1ubuntu0.6 [92.1 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libsasl2-modules-gssapi-mit amd64 2.1.27+dfsg-2ubuntu0.5 [643 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libsasl2-modules-db amd64 2.1.27+dfsg-2ubuntu0.5 [20.5 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libsasl2-2 amd64 2.1.27+dfsg-2ubuntu1.2 [53.8 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libldap-2.5-0 amd64 2.5.19+dfsg-0ubuntu0.22.04.1 [184 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9170 B]
Get:15 http://archive.ubuntu.com/ubuntu jammy/main amd64 libbrotli1 amd64 1.0.9-2build6 [315 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnghttp2-14 amd64 1.43.0-1ubuntu0.2 [76.9 kB]
```


10. En el AWS Management Console vamos a los grupos de seguridad de la instancia que acabamos de crear y modificamos las reglas de entrada para permitir el acceso a la instancia desde cualquier equipo, con esto ya podriamos iniciar Apache

Editar reglas de entrada

Las reglas de entrada controlan el tráfico entrante que puede llegar a la instancia.

ID de la regla del grupo de seguridad	Tipo	Protocolo	Intervalo de puertos	Origen	Descripción: opcional	Acción
sgr-055026fd34e0d1d37	HTTP	TCP	80	Persona...	0.0.0.0/0	Eliminar
sgr-03adb7641bf193dac	SSH	TCP	22	Persona...	0.0.0.0/0	Eliminar

[Agregar regla](#)

⚠ Las reglas cuyo origen es 0.0.0.0/0 o ::/0 permiten a todas las direcciones IP acceder a la instancia. Recomendamos configurar reglas de grupo de seguridad para permitir el acceso únicamente desde direcciones IP conocidas.

[Cancelar](#) [Previsualizar los cambios](#) [Guardar reglas](#)

11. Ponemos nuestra ip de EC2 en el navegador y debería de aparecer apache

Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in /usr/share/doc/apache2/README.Debian.gz**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/  
-- apache2.conf  
   |-- ports.conf  
-- mods-enabled  
   |-- *.load  
   |-- *.conf  
-- conf-enabled  
   |-- *.conf  
-- sites-enabled  
   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective `*-available/` counterparts. These should be managed by using our helpers `a2enmod`, `a2dismod`, `a2ensite`, `a2dissite`, and `a2enconf`, `a2disconf`. See their respective man pages for detailed information.
- The binary is called `apache2` and is managed using `systemd`, so to start/stop the service use `systemctl start apache2` and `systemctl stop apache2`, and use `systemctl status apache2` and `journalctl -u apache2` to check status. `system` and `apache2ctl` can also be used for service management if desired. **Calling `/usr/bin/apache2` directly will not work** with the default configuration.

Document Roots

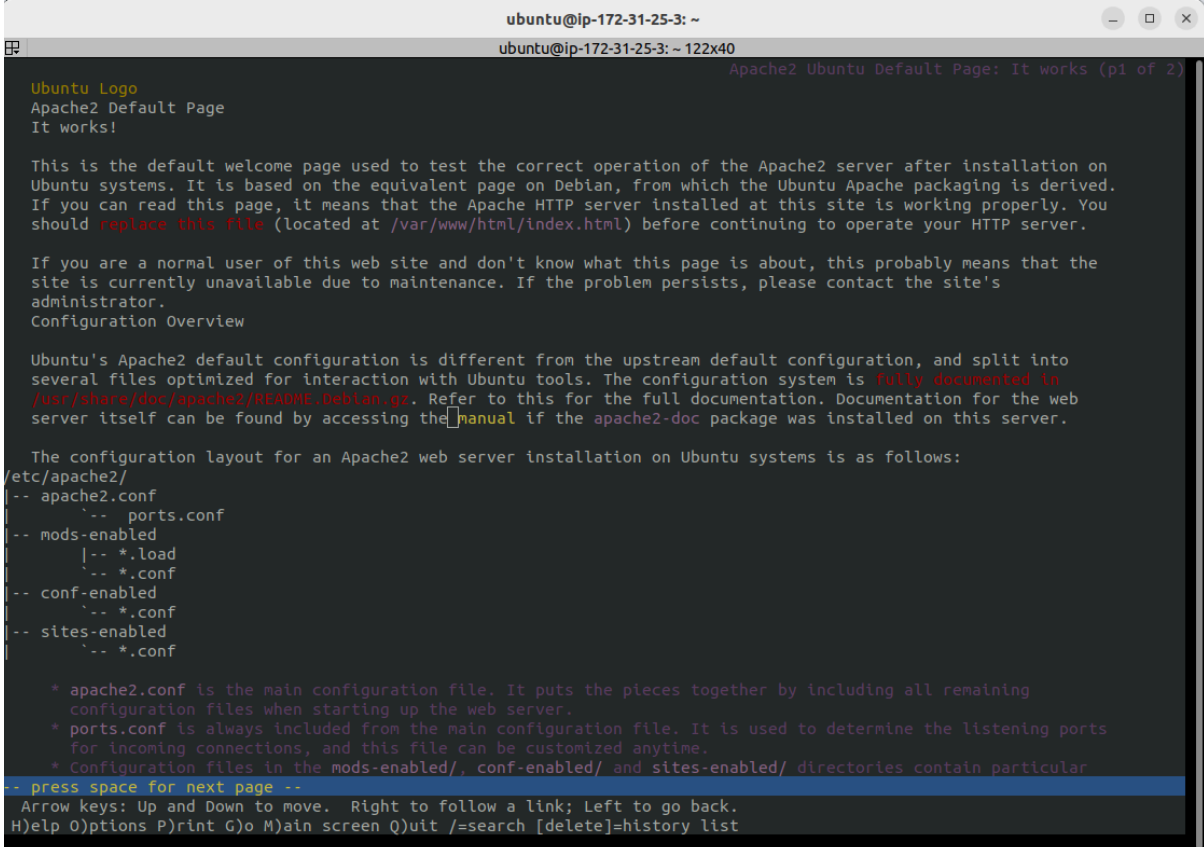
12. Para ver el apache dentro de la instancia EC2, primero nos instalamos un navegador de texto como Lynx

```
ubuntu@ip-172-31-25-3:~$ sudo apt install lynx -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  bzip2 lynx-common mailcap
```

13. Luego abrimos apache dentro de Lynx, con la ip local del EC2, poniendo localhost o en mi caso http://34.234.79.67.

```
ubuntu@ip-172-31-25-3:~$ lynx http://localhost
```

14. Debería aparecer esta pantalla, mostrándonos que funciona



```
ubuntu@ip-172-31-25-3: ~
ubuntu@ip-172-31-25-3: ~ 122x40
Apache2 Ubuntu Default Page: It works (p1 of 2)

Ubuntu Logo
Apache2 Default Page
It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on
Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived.
If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You
should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the
site is currently unavailable due to maintenance. If the problem persists, please contact the site's
administrator.
Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into
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|   |-- *.load
|   |-- *.conf
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|-- sites-enabled
|   |-- *.conf

* apache2.conf is the main configuration file. It puts the pieces together by including all remaining
  configuration files when starting up the web server.
* ports.conf is always included from the main configuration file. It is used to determine the listening ports
  for incoming connections, and this file can be customized anytime.
* Configuration files in the mods-enabled/, conf-enabled/ and sites-enabled/ directories contain particular
  configurations for various modules and sites.

-- press space for next page --
Arrow keys: Up and Down to move. Right to follow a link; Left to go back.
H)elp O)ptions P)rint G)o M)ain screen Q)uit /=search [delete]=history list
```

15. Ahora listamos los contenedores